Notes: Chetty, Looney & Kroft (AER, 2009)
Salience and Taxation: Theory and Evidence

Contents

1 Big picture

- **Question.** Do consumers respond less to taxes that are not salient, and how should tax incidence and excess burden be measured when taxpayers underreact to hidden taxes?
- **Empirical setting 1.** A field experiment in a grocery store posts tax-inclusive price tags for selected taxable products while leaving other products and nearby control stores unchanged.
- **Empirical setting 2.** A state-year panel of beer consumption compares the effect of beer excise taxes, which are included in posted prices, with general sales taxes, which are added at the register.
- **Main field-experiment result.** Posting tax-inclusive prices reduces demand for treated products by roughly 8 percent, almost as much as a comparable posted price increase would.
- **Main alcohol-tax result.** Increases in excise taxes reduce beer consumption by much more than equal increases in sales taxes.
- **Theory contribution.** The paper develops incidence and excess-burden formulas that allow the demand response to a tax to differ from the demand response to an equivalent price increase.
- **Core parameter.** The salience parameter θ or θ_τ measures the ratio of the tax elasticity of demand to the price elasticity of demand. Full optimization implies $\theta = 1$; inattention implies $\theta < 1$.
- **Main welfare lesson.** Economic incidence and deadweight loss depend on behavioral responses to the tax itself, not only on the price elasticity of demand. Statutory incidence can matter when salience differs across tax instruments.
- **Economic takeaway.** Tax design affects real behavior not only through rates and bases but also through presentation. Less salient taxes may raise more revenue with less immediate behavioral response, but welfare analysis must account for optimization errors.

2 Incremental contribution of the paper

- **Field-experiment contribution.** The paper provides a rare real-retail field experiment on tax salience, changing only posted information while holding actual tax rates fixed.
- **Two-method validation.** It combines an internally clean grocery-store experiment with an externally broader panel analysis of alcohol taxation. The two settings point to the same conclusion: less visible taxes generate smaller behavioral responses.
- **Tax-salience measurement.** It operationalizes salience as the ratio of the elasticity with respect to a hidden tax to the elasticity with respect to a posted price or posted-price tax.
- **Incidence extension.** The paper extends classical tax incidence formulas by allowing the shift in demand caused by a tax to be smaller than the shift caused by an equivalent price increase.
- **Welfare extension.** The paper modifies Harberger-style excess-burden formulas for cases in which the observed tax response does not reveal the full underlying price response of preferences.
- **Policy contribution.** It shows why transparency and salience are themselves tax-policy instruments. Tax-inclusive pricing, hidden fees, posted excise taxes, and register-added sales taxes can have different behavioral and welfare consequences even at the same statutory rate.
- **Behavioral public finance contribution.** The paper is an early benchmark for behavioral public economics: it links a behavioral bias directly to canonical objects in public finance–incidence, deadweight loss, and optimal tax design.

3 Data and measurement

3.1 Grocery-store field experiment

The field experiment was implemented in a supermarket in early 2006. In the store, posted shelf prices excluded a sales tax of 7.375 percent. The authors posted tags showing the tax-inclusive total price below the original pre-tax price tag for approximately 750 products in three taxable groups: cosmetics, hair care accessories, and deodorants.

The experiment uses weekly scanner data from week 1 of 2005 through week 13 of 2006. The treatment period is weeks 8–10 of 2006. The analysis is conducted at the category-store-week level, summing product-level quantities and revenues within product categories.

Interpretation: The treatment is informational, not fiscal. Actual after-tax prices are unchanged; only the visibility of the tax-inclusive price changes.

3.2 Treatment and control groups in the grocery design

The paper uses two control groups:

- **Within-store controls:** other taxable toiletries in the same aisles that did not receive tax-inclusive tags.
- **Control-store controls:** the same treatment and control product groups in two nearby stores where no tags were posted.

Interpretation: The within-store comparison controls for store-wide shocks, while the control-store comparison addresses product-category trends. Combining them produces the triple-difference estimate.

3.3 Survey data used to validate salience

The paper collects two surveys. A classroom survey shows students photos of the price tags and asks them to compute total bills. A grocery-store customer survey asks consumers which goods are taxable and asks them to report the local sales tax rate.

Interpretation: The surveys separate two mechanisms: consumers may not know the tax rate, or they may know the tax rate but fail to incorporate it while shopping. The grocery survey shows many consumers know tax status and tax rates reasonably well, favoring salience over ignorance.

3.4 Alcohol tax panel

The observational analysis uses annual state-level beer consumption and tax data from 1970 to 2003. Beer is useful because it faces two types of state tax:

- **Excise tax:** imposed per case, usually incorporated into the posted price.
- **Sales tax:** added at the register, less visible at the shelf.

The dependent variable is the change in log per-capita beer consumption. Key independent variables are changes in log gross-of-excise-tax price and log gross-of-sales-tax price. Controls include population, income, unemployment, alcohol regulation changes, year fixed effects, and region trends.

Interpretation: The beer panel tests whether salience matters in a stable, long-horizon setting rather than only in the short-run grocery intervention.

4 Empirical specifications

4.1 Salience parameter and demand equation

The paper writes the demand equation as

$$\log x(p, \tau^S) = \alpha + \beta \log p + \theta_\tau \beta \log(1 + \tau^S). \tag{1}$$

Here p is the posted pre-tax price, τ^S is the sales tax rate, and θ_τ measures how strongly consumers react to the tax relative to a price change.

- $\theta_\tau = 1$: full optimization; consumers treat a hidden tax like an equivalent price increase.
- $\theta_\tau = 0$: complete inattention to the sales tax.
- $0 < \theta_\tau < 1$: partial inattention or partial salience.

Interpretation: The key empirical object is not whether demand slopes down. It is whether the tax elasticity equals the price elasticity.

4.2 Grocery-store triple-difference design

The main grocery experiment estimates variants of

$$y_{cst} = \alpha_c + \alpha_s + \alpha_t + \delta(TreatmentStore_s \times TreatmentCategory_c \times TreatmentPeriod_t) + X'_{cst}\gamma + \varepsilon_{cst}, \tag{2}$$

where y_{cst} is quantity, revenue, or log quantity for category c, store s, and week t.

Interpretation: The coefficient δ is the effect of adding tax-inclusive tags in the treatment store for treated categories during the experimental period, net of category, store, and time shocks.

4.3 Estimating the salience parameter from the experiment

If posted tax-inclusive tags make the total price salient, then the difference between pre-tag and post-tag demand identifies the amount of underreaction to the register-added sales tax. The paper relates the effect of the tag to the category price elasticity:

$$(1 - \theta_\tau) = -\frac{\log x((1 + \tau^S)p, 0) - \log x(p, \tau^S)}{\varepsilon_{x,p} \log(1 + \tau^S)}. \tag{3}$$

Interpretation: When the tag effect is almost as large as the effect of an equivalent price increase, the implied θ_τ is close to zero: consumers were largely ignoring the hidden tax.

4.4 Beer-consumption panel specification

The state-year beer regression is

$$\Delta \log x_{jt} = \alpha + \beta \Delta \log(1 + \tau^E_{jt}) + \theta_\tau \beta \Delta \log(1 + \tau^S_{jt}) + X'_{jt}\rho + \varepsilon_{jt}, \tag{4}$$

where τ^E is the excise tax and τ^S is the sales tax.

Interpretation: Since the excise tax is included in posted prices, β is interpreted as the response to a salient tax-inclusive price change. The coefficient on the sales tax should be similar under full optimization, but it is much smaller empirically.

5 Theory: tax incidence and welfare with salience

5.1 Incidence with salience effects

Let demand be $D(p, t^S)$, where p is the pre-tax price and t^S is a unit sales tax. Supply is $S(p)$. The incidence on producers is

$$\frac{dp}{dt^S} = \frac{\partial D/\partial t^S}{\partial S/\partial p - \partial D/\partial p}. \tag{5}$$

The incidence on consumers is

$$\frac{dq}{dt^S} = 1 + \frac{dp}{dt^S}. \tag{6}$$

Interpretation: Relative to the standard model, the demand curve shifts by $t^S \partial D/\partial t^S$ rather than by $t^S \partial D/\partial p$. If consumers underreact to the tax, less of the tax is passed to producers.

5.2 Three incidence lessons

- **Producer incidence is attenuated by inattention.** If consumers barely respond to a hidden tax, the demand curve shifts little, so producers absorb less of the tax.
- **Tax elasticity and price elasticity are not interchangeable.** Two markets with the same tax elasticity can have different incidence if one has an elastic price response and low salience while the other has inelastic demand and high salience.
- **Statutory incidence matters.** A tax levied on the side where the tax is less salient can have different economic incidence than a tax levied on the other side.

5.3 Excess burden with salience

Let $\partial x^c/\partial p$ be the compensated price effect and $\partial x^c/\partial t^S$ the compensated tax effect. Define

$$\theta^c = \frac{\partial x^c/\partial t^S}{\partial x^c/\partial p}. \tag{7}$$

With fixed producer prices, the paper derives the approximation

$$EB(t^S) \simeq -\frac{1}{2}(t^S)^2 \theta^c \frac{\partial x^c}{\partial t^S}. \tag{8}$$

Equivalently, the deadweight loss is like the Harberger triangle for a perfectly salient tax of size $\theta^c t^S$.

Interpretation: If consumers completely ignore the tax in the relevant compensated sense, $\theta^c = 0$ and the incremental distortion is zero. But hidden taxes can still affect welfare through income effects, mistakes, and how revenue is used; the formula isolates the substitution distortion.

6 Identification strategy

6.1 Grocery field experiment

- The experiment changes salience while holding the actual tax rate fixed.
- Treated products are compared to untreated products in the same store and to the same product groups in nearby control stores.

- The identification assumption is common trends across treated and control categories absent the tag intervention.
- Placebo and permutation tests compare the treatment effect to thousands of possible false treatment assignments.
- The experimental concern is whether tags changed demand through salience or through novelty/Hawthorne effects; the beer tax panel addresses longer-run generalizability.

Interpretation: The grocery design has strong internal validity for the effect of posting tax-inclusive prices, but it is short-run and product-specific. The alcohol panel complements it with long-run policy variation.

6.2 Alcohol-tax design

- The alcohol analysis exploits state-year changes in beer excise taxes and sales taxes from 1970 to 2003.
- Excise taxes are included in posted prices; sales taxes are added at the register.
- Regressions include controls for business-cycle variables, alcohol regulations, year effects, and region trends.
- Robustness checks use an IV for excise-tax changes, three-year differences, ACCRA beer price data, food-exempt states, and ethanol-share outcomes.

Interpretation: The empirical test is simple: if consumers optimize fully, an equivalent excise-tax and sales-tax increase should have the same demand effect. The data strongly reject that equality.

7 Tables and figures

7.1 Exhibit 1: Tax-Inclusive Price Tags and Table 1: Survey Evidence

Read: Exhibit 1 shows the experimental price tag that displayed the total price including sales tax. Table 1 shows that only 18 percent of students computed tax-inclusive totals under the original tag, while 75 percent did so under experimental tags. Grocery customers generally knew which products were taxable and roughly knew the sales tax rate.

Interpretation: The survey evidence separates salience from ignorance. Consumers often know the tax but do not incorporate it at the moment of purchase unless the total price is made salient.

Economic message: The treatment works by changing attention to the total price, not by educating consumers about an unknown tax law.

EXHIBIT 1. TAX-INCLUSIVE PRICE TAGS

(a) Exhibit 1: Tax-inclusive price tags.

TABLE 1— SURVEY EVIDENCE: SUMMARY STATISTICS

	Mean	Median	Standard deviation
Panel A. Classroom survey			
Original price tags:			
Correct tax-inclusive price w/in \$0.25	0.18	0.00	0.39
Experimental price tags:			
Correct tax-inclusive price w/in \$0.25	0.75	1.00	0.43
t-test for equality of means: $p < 0.001$			
$N = 49$			
Panel B. Grocery store survey			
Local sales tax rate	7.48	7.39	0.80
(Actual rate is 7.375 percent)			
Fraction correctly reporting tax status			
All items	0.82	1.00	0.38
Beer	0.90	1.00	0.30
Cigarettes	0.98	1.00	0.15
Cookies	0.65	1.00	0.48
Magazines	0.87	1.00	0.34
Milk	0.82	1.00	0.38
Potatoes	0.81	1.00	0.39
Soda	0.76	1.00	0.43
Toothpaste	0.80	1.00	0.40
$N = 91$			

Notes: Panel A reports summary statistics for a survey of 49 students who were shown regular (non-tax-inclusive) price tax and the experimental (tax-inclusive) price tags. Statistics shown are for an indicator for whether individual reported total bill within 25 cents of total tax-inclusive price. See Web Appendix Exhibit 1 for survey instrument. Panel B reports summary statistics for a survey of 91 customers at the treatment grocery store. See Web Appendix Exhibit 2

(b) Table 1: Survey evidence.

7.2 Table 2: Grocery Experiment Summary Statistics and Table 3: DDD Analysis of Mean Quantity Sold

Read: Table 2 shows treatment and control categories are similar across stores. Table 3 shows the treatment-store DDD estimate is about -2.20 units per category per week, implying an approximate 8 percent demand reduction.

Interpretation: The main difference-in-differences estimate shows that making the tax-inclusive price visible reduces demand. The control-store panel shows no comparable change for the same product groups in other stores.

Economic message: Consumers behave as if the register-added sales tax is much less salient than the posted price.

TABLE 2— GROCERY EXPERIMENT: SUMMARY STATISTICS

	Treatment store		Control stores		Total
	Treatment products (1)	Control products (2)	Treatment products (3)	Control products (4)	All stores and products (5)
Panel A. Category-level statistics					
Weekly quantity sold per category	25.08 (24.1)	26.63 (38.1)	27.84 (27.4)	30.64 (47.0)	29.01 (42.5)
Weekly revenue per category	\$97.85 (81.9)	\$136.05 (169.9)	\$107.04 (92.3)	\$154.66 (207.7)	\$143.10 (187.1)
Number of categories	13	95	13	95	108
Panel B. Product-level statistics					
Pre-tax product price	\$4.46 (1.8)	\$6.26 (4.3)	\$4.52 (1.7)	\$6.31 (4.2)	\$6.05 (4.1)
Pre-tax product price (weighted by quantity sold)	\$4.27 (1.7)	\$5.61 (3.9)	\$4.29 (1.6)	\$5.59 (3.8)	\$5.45 (3.7)
Weekly quantity sold per product (conditional > 0)	1.47 (0.9)	1.82 (1.6)	1.61 (1.1)	1.98 (1.9)	1.88 (1.7)

Notes: Statistics reported are means with standard deviations in parentheses. Statistics are based on sales between 2005 week 1 and 2006 week 15. Data source is scanner data obtained from a grocery chain. The "treatment store" is the store where the intervention took place; the "control stores" are two nearby stores in the same chain. "Treatment products" are cosmetics, hair care accessories, and deodorants. "Control products" are other toiletries located in the same aisles; see Web Appendix Table 2 for complete list. Product price reflects actual price paid, including any discount if product

(a) Table 2: Grocery experiment summary statistics.

TABLE 3— EFFECT OF POSTING TAX-INCLUSIVE PRICES: *DDD* ANALYSIS OF MEAN QUANTITY SOLD

Period	Control categories	Treated categories	Difference
Panel A. Treatment store			
Baseline (2005:1–2006:6)	26.48 (0.22) [5,510]	25.17 (0.37) [754]	−1.31 (0.43) [6,264]
Experiment (2006:8–2006:10)	27.32 (0.87) [285]	23.87 (1.02) [39]	−3.45 (0.64) [324]
Difference over time	0.84 (0.75) [5,795]	−1.30 (0.92) [793]	$DD_{TS} = -2.14$ (0.68) [6,588]
Panel B. Control stores			
Baseline (2005:1–2006:6)	30.57 (0.24) [11,020]	27.94 (0.30) [1,508]	-2.63 (0.32) [12,528]
Experiment (2006:8–2006:10)	30.76 (0.72) [570]	28.19 (1.06) [78]	−2.57 (1.09) [648]
Difference over time	0.19 (0.64) [11,590]	0.25 (0.92) [1,586]	$DD_{CS} = 0.06$ (0.95) [13,176]
DDD Estimate			−2.20 (0.59) [19,764]

Notes: Each cell shows mean quantity sold per category per week, for various subsets of the sample. Standard errors

(b) Table 3: DDD mean quantity sold.

7.3 Table 4: Regression Estimates and Figure 1: Placebo Distribution

Read: Table 4 shows treatment effects around −2.20 to −2.27 units, −13.12 dollars of revenue, and −0.101 log quantity. Figure 1 shows the estimated log-quantity treatment effect lies in the left tail of 4,725 placebo estimates.

Interpretation: Table 4 strengthens the mean-comparison evidence using fixed effects and covariates. Figure 1 shows that the effect is unlikely under random reassignment of placebo treatment periods, stores, and categories.

Economic message: The field-experiment result is not a fragile artifact of one simple comparison.

TABLE 4—EFFECT OF POSTING TAX-INCLUSIVE PRICES: REGRESSION ESTIMATES

Dependent variable	Quantity per category (1)	Revenue per category ($) (2)	Log quantity per category (3)	Quantity per category (4)	Quantity (treat. categories only) (5)
Treatment	−2.20	−13.12	−0.101	−2.27	−1.55
	(0.60)	(4.89)	(0.03)	(0.60)	(0.35)
Average price	−3.15	−3.24		−3.04	−15.06
	(0.26)	(1.74)		(0.25)	(3.55)
Average price squared	0.05	0.06		0.05	1.24
	(0.00)	(0.03)		(0.00)	(0.34)
Log average price			−1.59		
			(0.11)		
Before treatment				−0.21	
				(1.07)	
After treatment				0.20	
				(0.78)	
Category, store, week FEs	x	x	x	x	x
Sample size	19,764	19,764	18,827	21,060	2,379

Notes: Standard errors, clustered by week, reported in parentheses. All columns report estimates of the linear regression model specified in equation (4). Quantity and revenue reflect total sales of products within a given category per week in each store. Average price is a weighted average of the prices of the products for sale in each category using a fixed basket of products (weighted by total quantity sold) over time. In column 3, observations are weighted by total revenue by category-store. Specification 4 includes "placebo" treatment variables (and their interactions) for the three-week period before the experiment and the three-week period after the experiment. Specification 5 reports DD estimates restricting the sample to treatment product categories only (at both treatment and control stores). In this specification, the "treat-

(a) Table 4: Regression estimates.

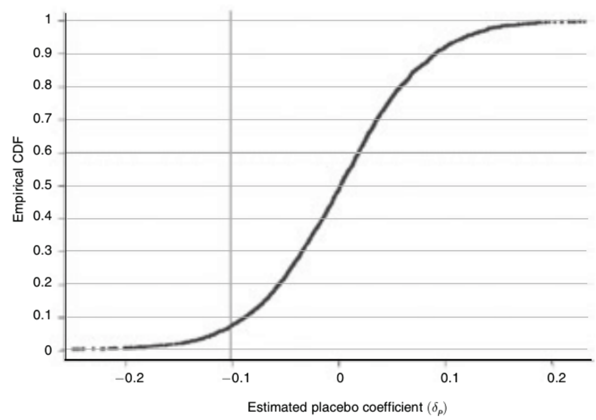


FIGURE 1. DISTRIBUTION OF PLACEBO ESTIMATES: LOG QUANTITY

Notes: This figure plots the empirical distribution of placebo effects (G) for log quantity. The CDF is constructed from 4,725 estimates of δ_p using the specification in column 3 of Table 4. No parametric smoothing is applied: the CDF appears smooth because of the large number of points used to construct it. The vertical line shows the treatment effect estimate reported

(b) Figure 1: Placebo distribution.

7.4 Table 5: Beer Consumption, Taxes, and Regulation Summary Statistics

Read: Table 5 reports 1,666 state-year observations. Average per-capita beer consumption is 243.2 cans. Average state excise tax is 6.5 percent of price, and average sales tax is 4.3 percent.

Interpretation: The beer setting provides independent long-run variation in salient versus less salient taxes.

Economic message: The alcohol panel tests whether salience matters outside the short-run grocery experiment.

Table 5— Summary Statistics for State Beer Consumption, Taxes, and Regulation

Per capita beer consumption (cans)	243.2
	(46.1)
State beer excise tax ($/case)	0.51
	(0.50)
State beer excise tax (percent)	6.5
	(8.2)
Sales tax (percent)	4.3
	(1.9)
Drinking age is 21	0.73
	(0.44)
Drunk driving standard	0.65
	(0.47)
Any alcohol regulation change	0.19
	(0.39)
N (number of state-year pairs)	1,666

Notes: Statistics reported are means with standard deviations in parentheses. Observations are by state for each year from 1970 to 2003. "Drinking age is 21" is an indicator for whether the state-year has a legal drinking age of 21. "Drunk driving standard" indicates state-year has a threshold blood alcohol content level above which one is automatically guilty of drunk driving. "Any alcohol regulation change" is a dummy variable equal to one in any year where a state has raised the drinking age or implemented a stricter drunk driving standard, an administrative license revocation law, or a zero tolerance youth drunk driving law. See Web Appendix A for data sources and sample definition.

Table 5: Summary statistics for state beer consumption, taxes, and regulation.

7.5 Figure 2A and Figure 2B: Beer Consumption and Excise versus Sales Taxes

Read: Figure 2A shows a clear negative relation between changes in excise-tax-inclusive prices and beer consumption. Figure 2B shows a much flatter relation for sales tax changes.

Interpretation: The visual evidence previews the regression results: salient excise taxes shift consumption, while less salient sales taxes produce much smaller changes.

Economic message: Taxes included in posted prices appear to matter more for real behavior than taxes added later at checkout.

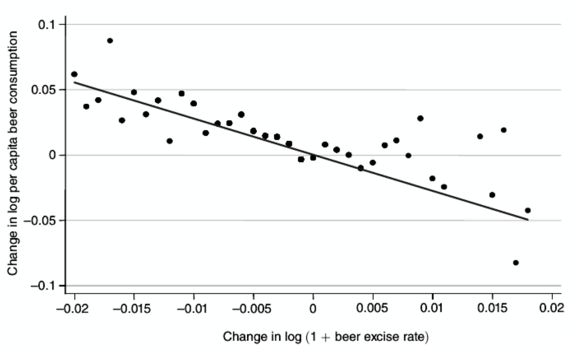


Figure 2A. Per Capita Beer Consumption and State Beer Excise Taxes

(a) Figure 2A: Beer consumption and excise taxes.

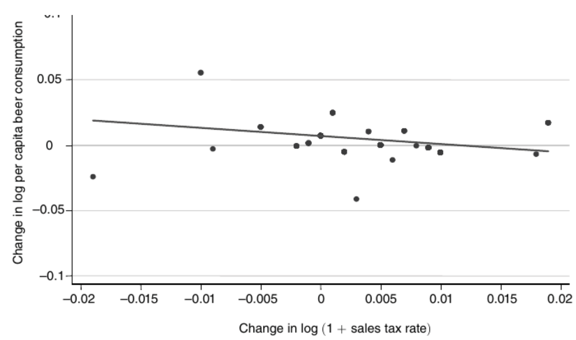


Figure 2B. Per Capita Beer Consumption and State Sales Taxes

Notes: These figures plot within-state annual changes in beer consumption against within-sta changes in gross-of-tax-prices ($1 + t^E$ and $1 + t^S$). To construct Figure 2A, we round each sta excise tax change to the nearest tenth of a percent (0.1 percent), and compute the mean change i log beer consumption for observations with the same rounded excise tax change. Figure 2A plo

(b) Figure 2B: Beer consumption and sales taxes.

7.6 Table 6: Beer Consumption Regressions and Table 7: Robustness Checks

Read: Table 6 estimates excise-tax elasticities around −0.88 to −0.91 in the main specifications, while sales-tax effects are close to zero after controls. Table 7 shows the result is robust to IV for excise-tax policy changes, three-year differences, ACCRA beer prices, food-exempt states, and ethanol-share outcomes.

Interpretation: These tables are the external-validity counterpart to the field experiment. A tax included in posted prices has an order-of-magnitude larger effect on consumption than a tax added at the register.

Economic message: The degree of salience is a stable determinant of tax response, not merely a short-run retail-display artifact.

TABLE 6— EFFECT OF EXCISE AND SALES TAXES ON BEER CONSUMPTION

	Baseline (1)	Business cycle (2)	Alcohol regulations (3)	Region trends (4)
Dependent variable: Change in log (per capita beer consumption)				
Δ log (1 + excise tax rate)	−0.88	−0.91	−0.89	−0.71
	(0.17)	(0.17)	(0.17)	(0.18)
Δ log (1 + sales tax rate)	−0.20	−0.01	−0.02	−0.05
	(0.30)	(0.30)	(0.30)	(0.30)
Δ log (population)	0.03	−0.07	−0.07	−0.09
	(0.06)	(0.07)	(0.07)	(0.08)
Δ log (income per capita)		0.22	0.22	0.22
		(0.05)	(0.05)	(0.05)
Δ log (unemployment rate)		−0.01	−0.01	−0.01
		(0.01)	(0.01)	(0.01)
Alcohol regulation controls			x	x
Year fixed effects	x	x	x	x
Region fixed effects				x
F-test for equality of tax elasticities (prob > F)	0.05	0.01	0.01	0.06
Sample size	1,607	1,487	1,487	1,487

Notes: Standard errors, clustered by state, in parentheses. All specifications are estimated on full sample for which data are available (state unemployment rate data are unavailable in early years). Column 3 includes three indicators for whether the state implemented per se drunk driving standards, administrative license revocation laws, or zero tolerance youth

(a) Table 6: Beer tax estimates.

TABLE 7— EFFECT OF EXCISE AND SALES TAXES ON BEER CONSUMPTION: ROBUSTNESS CHECKS

	Dependent variable: Change in log (per capita beer consumption)				Dep. var.
	IV for excise w/ policy (1)	3-Year differences (2)	IV for ACCRA beer price (3)	Food exempt (4)	Share ethanol from beer (5)
Δ log (1 + excise tax rate)	−0.63	−1.10		−0.91	0.16
	(0.28)	(0.47)		(0.22)	(0.13)
Δ log (beer price)			−0.88		
			(0.42)		
Δ log (1 + sales tax rate)	−0.03	−0.00	0.10	−0.14	0.25
	(0.30)	(0.33)	(0.59)	(0.30)	(0.22)
Δ log (population)	−0.06	−1.24	−0.06	0.03	0.09
	(0.07)	(0.33)	(0.15)	(0.07)	(0.05)
Δ log (income per capita)	0.22	0.08	0.23	0.22	0.01
	(0.05)	(0.05)	(0.10)	(0.05)	(0.03)
Δ log (unemployment rate)	−0.01	−0.00	0.00	−0.01	0.00
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Alcohol regulation controls	x	x	x	x	x
Year fixed effects	x	x	x	x	x
F-test for equality of tax elasticities (prob > F)	0.15	0.05	0.12	0.04	0.73
Sample size	1,487	1,389	825	937	1,487

Notes: Standard errors, clustered by state, in parentheses. Column 1 replicates column 3 of Table 6, instrumenting for excise tax rate changes with the nominal excise tax rate divided by the average price of a case of beer from 1970 to 2003

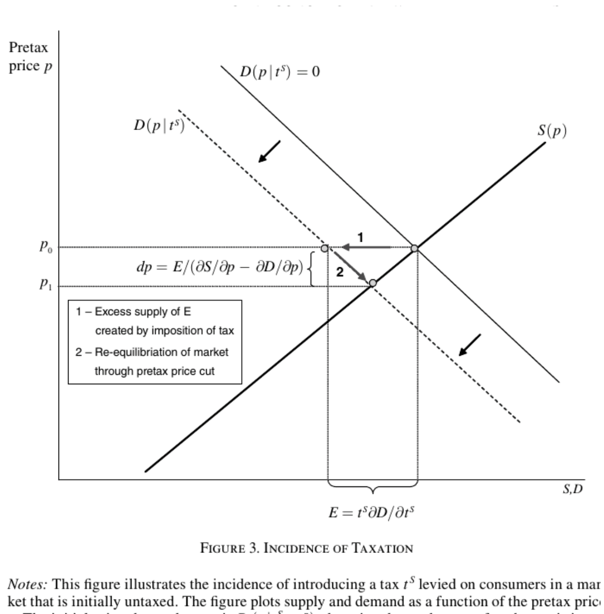
(b) Table 7: Robustness checks.

7.7 Figure 3: Incidence of Taxation and Figure 4: Excess Burden with No Income Effect

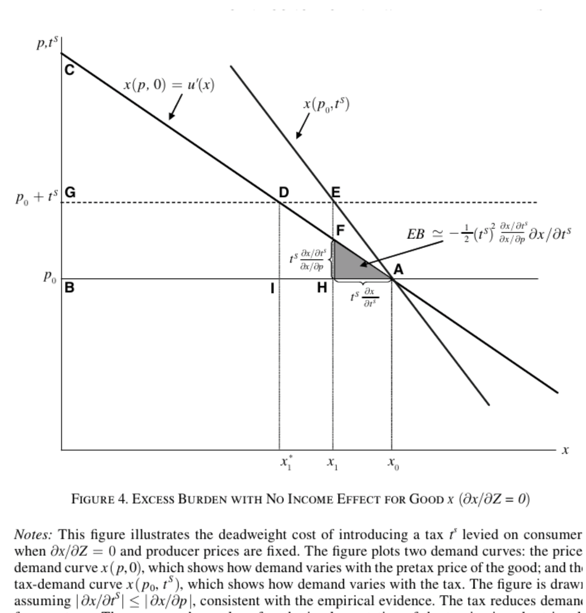
Read: Figure 3 modifies the standard incidence diagram by making the demand shift depend on $\partial D/\partial t^S$ rather than $\partial D/\partial p$. Figure 4 shows excess burden when the tax-demand curve is flatter than the price-demand curve.

Interpretation: The theory figures show why tax salience matters for public finance fundamentals. Incidence and deadweight loss cannot be computed from the price elasticity alone when tax and price responses differ.

Economic message: Behavioral underreaction changes both who bears a tax and how costly the tax is.



(a) Figure 3: Incidence of taxation.



(b) Figure 4: Excess burden with no income effect.

8 Interpretive synthesis

- **Empirical synthesis.** The grocery experiment and alcohol panel use very different identification strategies but reach the same conclusion: consumers underreact to taxes that are not included in posted prices.
- **Theoretical synthesis.** The paper does not simply document a behavioral anomaly; it shows how canonical public-finance formulas change when the tax elasticity differs from the price elasticity.
- **Policy synthesis.** Tax salience is a policy variable. Governments can raise revenue with hidden taxes, but lower behavioral responses do not automatically imply lower welfare costs under all objectives.
- **Conceptual distinction.** Inattention to taxes is not the same as inelastic demand. Both can reduce observed quantity responses, but they have different implications for incidence and excess burden.
- **Research-design insight.** Combining a field experiment with observational tax variation makes the evidence stronger than either approach alone.

9 Short summary

- The paper studies whether consumers respond less to non-salient taxes.
- In a grocery-store experiment, tax-inclusive tags reduce demand by about 8 percent.
- Survey evidence shows consumers generally know tax status but do not fully incorporate taxes when prices exclude them.
- In state beer data, excise taxes reduce consumption much more than sales taxes.
- The implied salience parameter is far below one for register-added taxes.
- The paper extends tax incidence and excess-burden formulas to allow tax and price elasticities to differ.
- Less salient taxes are borne less by producers and generate different efficiency costs than equally sized salient taxes.
- The incremental contribution is to connect behavioral salience evidence directly to the core formulas of public economics.

10 Conclusion

10.1 Core conclusion

Consumers underreact to taxes that are not included in posted prices. This underreaction is visible both in a field experiment that makes sales taxes salient and in state-level alcohol data comparing salient excise taxes with less salient sales taxes.

10.2 Economic intuition

Consumers often focus on the posted price while shopping. A hidden tax changes the true total price but does not shift perceived marginal cost one-for-one. Therefore demand responds less to the tax than to an equivalent posted price increase.

10.3 Incremental contribution in one sentence

Chetty, Looney, and Kroft transform tax salience from a behavioral observation into a quantitative public-finance object by estimating tax-price underreaction and embedding it into incidence and welfare formulas.

10.4 Takeaway

Tax policy depends not only on rates and bases, but also on visibility. Public finance models and policy evaluations must distinguish price elasticities from tax elasticities when taxes are not fully salient.